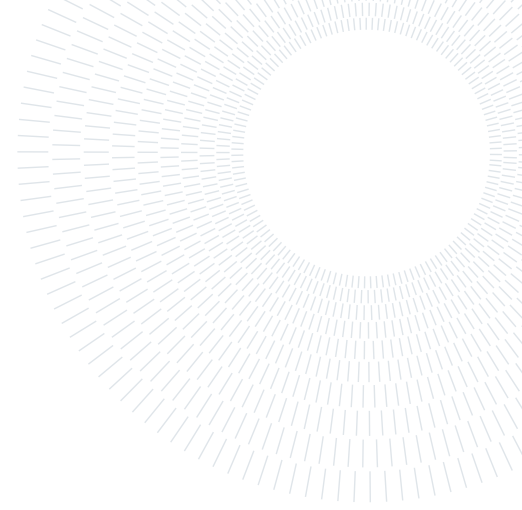




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Bayesian modeling of transconjugation efficacy under chemical stressors

BAYESIAN STATISTICS COURSE - MATHEMATICAL ENGINEERING

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Abstract: This work studies the effect of chemical stressors on bacterial transconjugation efficacy through a hierarchical Bayesian modeling approach. The transconjugation rate, defined as the ratio of transconjugant to donor counts, is estimated accounting for dilution factors and measurement uncertainty.

Transconjugant counts are modeled using generalized linear models with a log link under Poisson and Negative Binomial assumptions, while donor counts are incorporated as exposure terms and modeled hierarchically to propagate their variability. The experimental design includes combinations of temperature, ibuprofen, and dimethyl sulfoxide (DMSO) concentrations, with biological replicates handled through random effects.

Model performance is evaluated via posterior predictive checks and information criteria (LOO), and shrinkage priors are used to control complexity.

Results show a consistent effect of DMSO and an interaction between ibuprofen and temperature. Substantial replicate-level variability supports the hierarchical approach, which enables coherent uncertainty quantification across all model levels.

Key-words: Bayesian hierarchical generalized linear models; Chemical stressors; Experimental design; Shrinkage priors; Transconjugation rate

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1. Introduction

The main goal of this project is to model the frequency of transconjugation under different chemical stressors using a Bayesian approach.

1.1. Trasconjugation

Transconjugation is the ability of a bacterium to receive genetic material, a plasmid, from another bacterium through conjugation. This horizontal gene transfer mechanism contributes to the evolution and spread of various characteristics within bacterial populations, including traits such as antibiotic resistance. Figure 1 illustrates the phenomenon.

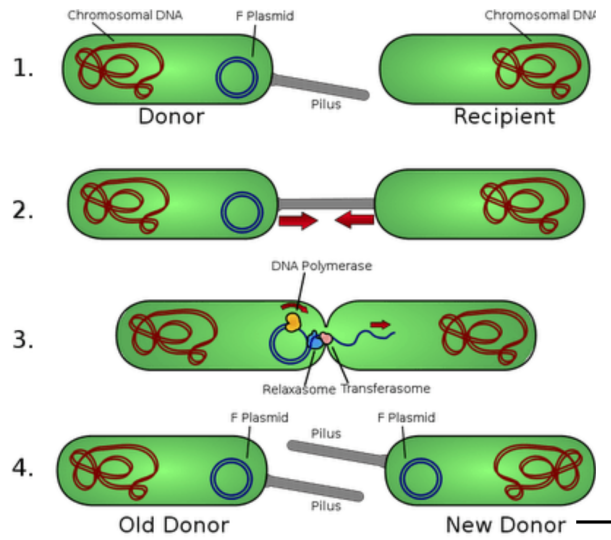


Figure 1: Bacterial conjugation and plasmid transfer (transconjugation) between donor and recipient cells.

Source: Laboratory protocol

This biological mechanism motivates the need for a statistical model capable of capturing variability across experimental conditions and replicates.

1.2. Bacterial Count

Estimating the frequency of transconjugation requires the calculation of the transconjugation rate, defined as the ratio between the number of transconjugant bacteria and the number of donor bacteria. Obtaining reliable bacterial counts is a non-trivial task, as bacteria are initially suspended in liquid solutions and cannot be counted directly.

To perform bacterial counting, a small volume of the bacterial suspension is taken and plated onto nutrient-rich agar. After an incubation period, viable bacteria reproduce and form visible colonies on the plate, which are considerably easier to count than individual cells. Bacterial quantification is therefore based on counting colonies rather than single bacteria, under the assumption that each colony originates from one viable bacterial cell.

In order to simplify the counting process, the original solution is typically diluted before plating. Dilution reduces the concentration of bacteria in the plated sample, resulting in fewer and more clearly separated colonies. The

dilution factor is then used to infer the bacterial concentration in the original solution.

This procedure, however, presents several limitations. Errors may be introduced during the dilution process and inaccuracies in dilution factors can affect the final count. In addition, when bacterial concentrations are high, colonies may overlap, leading to underestimation of the true number of bacteria. These sources of uncertainty must be considered when computing transconjugation frequencies.

These sources of uncertainty motivate a hierarchical Bayesian treatment in which donor and transconjugant colony counts are jointly modeled, allowing uncertainty from dilution and counting to propagate to inference on the transconjugation rate.

2. Experimental Design

The first step of the study consisted in a detailed understanding of the experimental design and conditions that generated the dataset. The entire workflow is based on the repetition of a set of experiments under different conditions. First, the general experimental setup is described and all components and factors involved are examined to allow for a correct interpretation of the data. Subsequently, the organization of experimental replicates is illustrated.

The experiment involves two strains of *E. coli* bacteria:

- CL218: donor strain resistant to Cefotaxime (CAZ);
- J53: recipient strain resistant to Azide.

The two strains are plated on LB agar (LA) supplemented with different combinations of selective agents (LA + AZIDE, LA + CAZ, LA + CAZ + AZIDE).

These selective conditions allow discrimination between donor, recipient, and transconjugant populations based on their antibiotic resistance profiles, ensuring that only the relevant bacterial populations are considered in subsequent analyses.

After an incubation period of 4 hours at 37°C, a volume of 500 μ l is collected from the CL218 culture grown on LA + CAZ plates and from the J53 culture grown on LA + AZIDE plates. These suspensions are then distributed into six different flasks. Three flasks correspond to the control condition, while the remaining three correspond to the treatment condition. In addition, a separate tube containing only donor bacteria J53 is prepared and used to estimate donor cell counts. The entire process is shown in Figure 2.

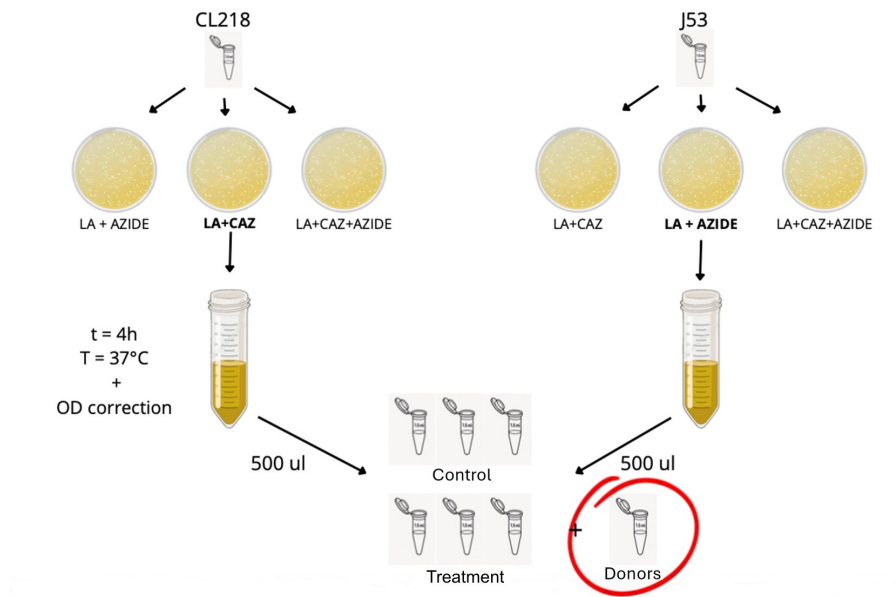


Figure 2: Experimental workflow

Source: Laboratory protocol

The six flasks containing both donors and recipients are used to quantify transconjugants, as described in Section 1. Specifically, a small volume from each flask is plated onto agar containing both CAZ and AZIDE. Under these selective conditions, only recipient cells that have acquired the plasmid from the donor strain, and therefore resistance to CAZ, are able to grow and are counted as transconjugants.

For each of the six flasks, bacterial counts are performed three times using serial dilutions with dilution factors of 10^{-1} , 10^{-2} and 10^{-3} . Similarly, donor-only counts are repeated three times using higher dilution factors of 10^{-6} , 10^{-7} , and 10^{-8} to account for their higher concentration.

The entire experimental procedure is repeated under different combinations of chemical stressors, defined ac-

according to the design of experiment shown in Figure 3. The three factors considered are:

- temperature, taking values of 28°C or 40°C;
- ibuprofen (IBU) concentration, taking values of 0, 6.25, or 50 mg/l;
- dimethyl sulfoxide (DMSO) concentration, taking values of 0 %v/v or 10 %v/v.

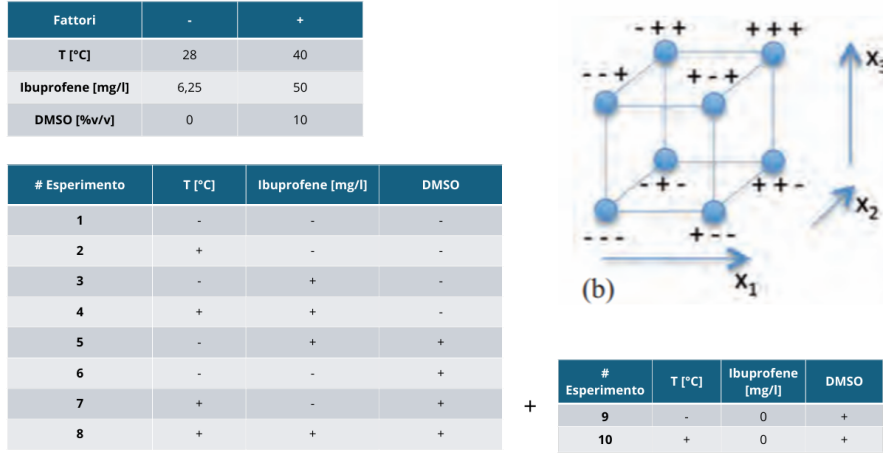


Figure 3: Design of Experiment

Source: Laboratory protocol

For each experiment, a specific combination of the three factors defines the treatment condition of the three treatment flasks. For the corresponding control flasks, one of the factors is set to zero in order to evaluate the effect of its presence or absence. In particular, in experiments 1–8 the ibuprofen concentration is set to zero in the control condition, while in experiments 9 and 10 the DMSO concentration is set to zero, given that ibuprofen is already absent.

For each experimental condition reported in Table 3, the experiment is repeated three times under identical conditions, corresponding to three biological replicates, with the exception of experiments 9 and 10, which are repeated only twice.

Experiments 5 and 6 are conducted simultaneously and share the same three control flasks (and likewise experiments 7 and 8), implying that comparisons involving these controls are not independent: a control flask deviation impacts multiple experiments at once. In addition, counts are collected across dilutions within each biological replicate, and replicates are collected within each treatment condition of the factorial design.

These nested and shared components motivate a hierarchical Bayesian model that explicitly represents replicate and control level variability (Section 4).

3. Data and exploratory analysis

Two datasets are analysed. The first one contains the trasconjugants counts for each replicate of the experiments and the second one contains the donors counts for each replicate of the experiments.

The columns specify:

- the experiment index $\in \{1, \dots, 10\}$
- the experiment replica index $\in \{1, 2, 3\}$
- the dilution factor $\in \{-1, -2, -3\}$ or $\in \{-6, -7, -8\}$
- the control or treatment condition $\in \{0, 1\}$
- Temp value $\in \{28, 40\}$
- IBU concentration value $\in \{0, 6.25, 50\}$
- DMSO concentration value $\in \{0, 10\}$
- the count $\in \mathbb{Z}^+$

The datasets initially contain 450 and 72 measurements respectively. Some counts in the first dataset are reported as TMTC, which means "too much to count", so they are removed from the final dataset, resulting in a final sample size of 378 measurements.

The preprocessing consists of normalizing the IBU, DMSO and temperature variables, so that they have zero mean and unit variance.

Finally, the experiment indeces are reassigned $\in \{1, \dots, 8\}$, so that experiments 5 and 6 and experiments 7 and 8 are considered together, as explained in the experiment workflow in Section 2.

An exploratory analysis was performed separately for the transconjugants and donors datasets to characterize the distribution of the counts and to inspect the effect of the experimental factors.

3.1. Descriptive statistics

Tables 1 and 2 report the summary statistics of the final count variable across experimental conditions.

Table 1 shows the average final count and corresponding standard deviation for each combination of IBU concentration, DMSO percentage and temperature in the donor dataset. The largest mean values are observed for the control condition (IBU = 0) with DMSO = 10 at both temperatures, while higher IBU concentrations generally correspond to lower counts, particularly at 28°C.

Table 2 reports the same experimental conditions for the second dataset, together with the average donor count and the $\log_{10}$ ratio between final and donor counts. The ratio is negative in all conditions, indicating that the final counts are consistently lower than the corresponding donor means. More negative values are observed in the absence of IBU combined with DMSO = 10, especially at 28°C, whereas milder reductions are found for intermediate IBU levels.

Mean counts by experiment and control: Figure 4 shows the mean real count computed for each combination of experiment index, replica and control condition. The means were obtained by grouping the data by experiment, replica and control, and averaging the corresponding counts. Each point in the plot therefore represents the average count for a specific replicate within a given experiment and condition.

Table 1: Summary statistics for IBU, DMSO and Temperature conditions.

i	IBU	DMSO	Temperature	Final Count (Avg)	St. Dev.
1	6.25	0	28	7.6556e+07	7.2507e+07
2	6.25	0	40	1.1167e+08	6.1514e+07
3	50.00	0	28	3.3444e+07	2.7354e+07
4	50.00	0	40	1.4256e+08	1.4124e+08
5	50.00	10	28	3.7444e+07	3.2481e+07
6	50.00	10	28	3.7444e+07	3.2481e+07
7	50.00	10	40	5.1111e+07	3.5775e+07
8	50.00	10	40	5.1111e+07	3.5775e+07
9	0.00	10	28	2.2867e+08	1.8726e+08
10	0.00	10	40	2.2867e+08	1.8726e+08

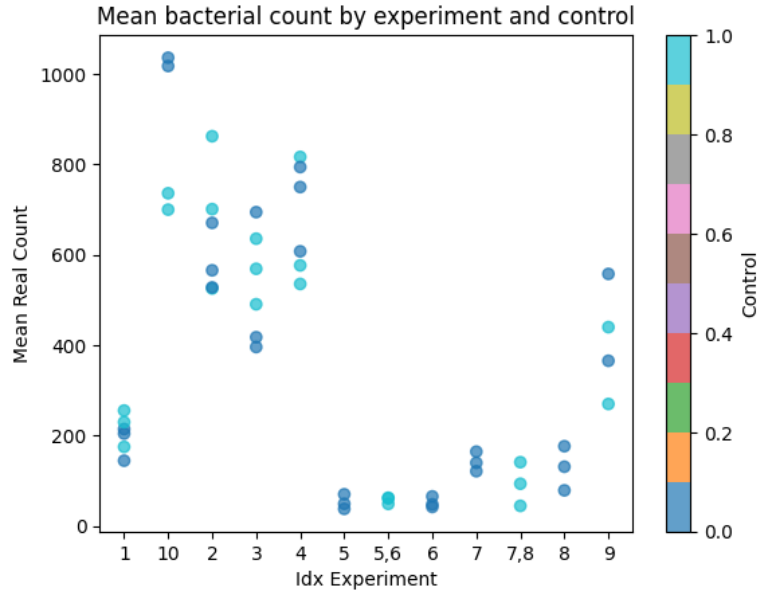


Figure 4: Mean real count by experiment and control condition. Each point represents the mean count for one replicate within an experiment. Colors indicate control (0) or treatment (1).

The figure highlights a substantial variability across experiments. Experiments 2, 3 and 4 show markedly higher mean counts compared to experiments 5, 6 and 7, where the averages are considerably lower. A separation between control and treatment is visible in some experiments, with treatment points often associated with reduced mean counts, although this pattern is not uniform across all experimental indices. The substantial variability across experiments and replicates suggests incorporating replicate-specific random effects within a hierarchical modeling framework.

3.2. Joint effect of temperature, IBU and DMSO.

To explore the combined effect of temperature, IBU and DMSO on the bacterial counts, a pivot table was computed by averaging the real counts for each combination of Temp and DMSO (rows) and IBU (columns). The resulting means are shown in Figure 5 as a heatmap.

Table 2: Summary statistics across experimental conditions. The ratio is expressed in $\log_{10}$ scale.

IBU	DMSO	Temp	Final Count (Avg)	Std. Dev,	Donor Count (Avg)	Ratio ($\log_{10}$)
0.00	0.0	28.0	3122.9166	1965.9908	9.8416e+07	-4.6125
0.00	0.0	40.0	1062.0833	863.9574	1.5250e+08	-5.1578
0.00	10.0	28.0	560.5555	626.6445	2.2866e+08	-5.6170
0.00	10.0	40.0	1201.1111	1402.4174	2.2866e+08	-5.2809
6.25	0.0	28.0	6644.4444	3996.2241	7.6555e+07	-4.0655
6.25	0.0	40.0	1177.7778	711.4657	1.1166e+08	-4.9861
6.25	10.0	28.0	3840.7407	4633.3446	3.7444e+07	-3.9891
6.25	10.0	40.0	4307.4074	4444.6441	5.1111e+07	-4.0744
50.00	0.0	28.0	1138.1481	433.9310	3.3444e+07	-4.5081
50.00	0.0	40.0	1046.6666	846.0212	1.4255e+08	-5.1506
50.00	10.0	28.0	3670.3703	4770.2816	3.7444e+07	-4.0088
50.00	10.0	40.0	4588.8888	4248.9057	5.1111e+07	-4.0473

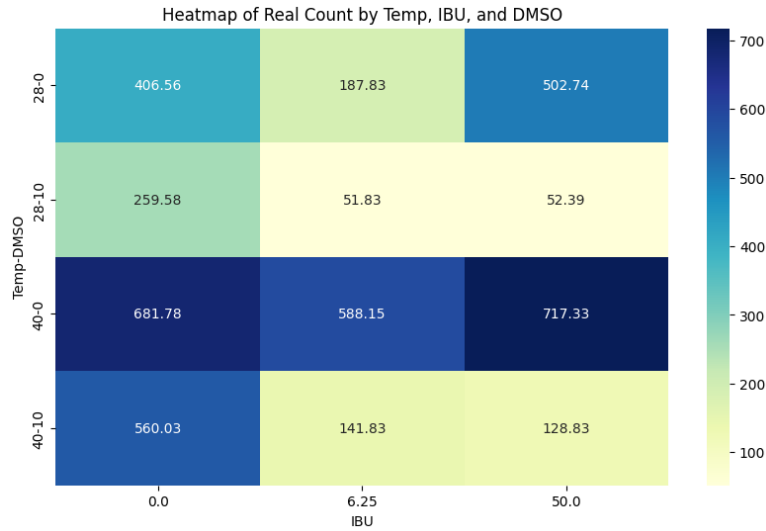


Figure 5: Heatmap of mean real count by temperature, DMSO and IBU. Each cell reports the average count for a specific combination of experimental conditions.

The heatmap reveals clear differences across temperature levels. At DMSO = 0, the highest mean counts are observed, particularly at Temp = 40, where values exceed 550. In contrast, the presence of DMSO = 10 is associated with a strong reduction in counts across all IBU levels, with mean values close to 50 in some configurations.

For IBU = 6.25, mean counts are consistently lower compared to IBU = 0 and IBU = 50, suggesting a possible non-linear effect of IBU concentration. Overall, the figure highlights interaction patterns between temperature and DMSO, as the effect of IBU appears to depend on the combination of these two factors.

These non-additive patterns suggest interaction effects among the stressors, motivating the inclusion of interaction terms in the hierarchical regression model.

4. Methodology

A Bayesian framework is exploited to model the response surface of trasconjugacy efficacy. The first step consists in the construction of an hierarchical generalized linear model which is able to capture the underlying phenomenon and the variability introduced by experimental conditions and replicates.

The general structure underneath the model is reported in Figure 6. Trasconjugants counts in both control and treatment condition are modeled as a function of control covariates and treatment covariates. The random effect associated with experimental replicates is shared across the corresponding counts.

Donor counts are used as proxies for measurement errors and incorporated in the Bayesian model as the offset term in the computation of the transconjugation rate. Rather than being treated as fixed quantities, donor counts are modeled within the Bayesian framework to account for their variability and measurement uncertainty.

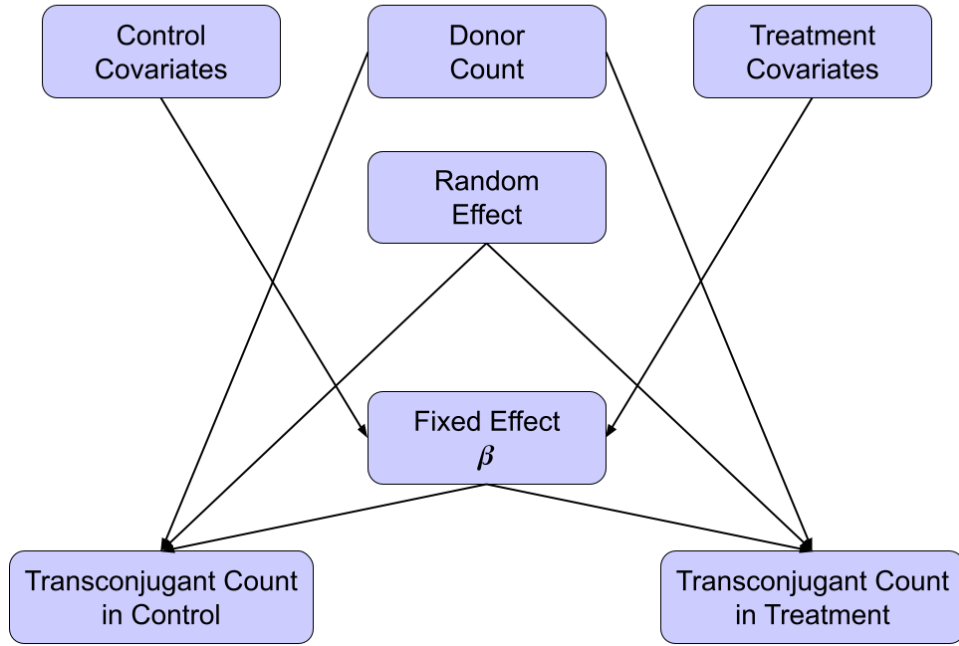


Figure 6: Graphical representation of the hierarchical Bayesian generalized linear model. Control and treatment covariates enter the fixed effect term through the coefficient vector β , while the random effect accounts for replicate-level variability. Donor counts are incorporated as an offset in the computation of the transconjugation rate.

4.1. Hierarchical Generalized Linear Model

Since trasconjugant counts are non-negative integer values, the natural modeling framework is based on discrete count distributions. The canonical choice is the Poisson distribution. In addition, the Negative Binomial is also considered as an alternative specification to account for the extra variability in the real experimental counts.

Thus, these two likelihood families provide two alternative models for the trasconjugant counts T . With the appropriate indexing, the two competing models are defined as follows in (1) and (3)

$$T_{i,j,k,c} \mid \lambda_{i,j,c}, X_{i,c}, \rho_k \stackrel{\text{ind}}{\sim} \text{Poisson}(\rho_k \lambda_{i,j,c}) \quad (1)$$

$$\lambda_{i,j,c} = \alpha_i \theta_{i,j,c} \quad (2)$$

where:

- $i \in \{1, \dots, 8\}$ = experiment
- $j \in \{A, B, C\}$ = experiment replica
- $k \in \{1, \dots, 9\}$ = observations within each experiment replica
- $c \in \{0, 1\}$ = control (0) or treatment (1)
- $T_{i,j,k,c}$ = transconjugant count
- $X_{i,c}$ = vector containing covariates (1, Temp, [IBU], [DMSO])
- $\rho_k = 10^{-\lceil \frac{k}{3} \rceil}$ = dilution factor
- α_i = donor count (exposure) for experiment i
- $\theta_{i,j,c}$ = transconjugation rate per donor

$$T_{i,j,k,c} \mid \lambda_{i,j,c}, X_{i,c}, \rho_k, \phi_T \stackrel{\text{ind}}{\sim} \text{NB}(\rho_k \lambda_{i,j,c}, \phi_T) \quad (3)$$

$$\lambda_{i,j,c} = \alpha_i \theta_{i,j,c} \quad (4)$$

where the following parametrization has been used:

- $\rho_k \lambda_{i,j,c}$ = average observed counts,
- ϕ_T = overdispersion parameter.

4.1.1 Linear predictor

The transconjugacy rate is expressed as a linear function of the parameters β representing the effect of Temp, [IBU] and [DMSO], plus a random effect $b_{i,j}$ accounting for variability between experimental replicates that is not explained by the chemical stressors. The offset α_i accounts for the number of donors in experiment i and it is modeled within a Bayesian framework in (8). A log link function is used to introduce the linear predictor as shown in (5).

$$\log(\lambda_{i,j,c}) = \log(\alpha_i) + b_{i,j} + \beta^\top F(\mathbf{X}_{i,c}) \quad (5)$$

where:

- $b_{i,j}$ = random effect $b_{i,j}$ for experiment i and replica j ;
- β = vector containing parameters accounting for the covariates effect;
- F = function applied to the covariates in $X_{i,c}$, the covariate vector for experiment i under condition c .

4.1.2 Bayesian model for donors

Rather than treating donor counts as fixed exposure quantities, we model them hierarchically so as to propagate measurement uncertainty into the estimation of the transconjugation rate. In line with the specification adopted for transconjugant counts, two alternative likelihood specifications are considered for donor counts, namely a Poisson and a Negative Binomial distribution, allowing us to assess the impact of potential overdispersion. For the first model, the mean of the Poisson is proportional to the true donor count α_i of experiment i , scaled by the corresponding dilution factor.

$$D_{i,k} \mid \alpha_i, \delta_k \stackrel{\text{ind}}{\sim} \text{Poisson}(\delta_k \alpha_i) \quad (6)$$

where:

- $D_{i,k}$ = donor count measurement for experiment i at dilution level k ;
- $\delta_k = 10^{-5 - \lceil \frac{k}{3} \rceil}$ dilution factor.

The second model is:

$$D_{i,k} \mid \alpha_i, \delta_k, \phi_D \stackrel{\text{ind}}{\sim} \text{NB}(\delta_k \alpha_i, \phi_D) \quad (7)$$

where:

- $\delta_k \alpha_i$ = average observed counts,
- ϕ_D = overdispersion parameter.

Donor and transconjugant counts are assumed to be conditionally independent given the latent transconjugation rate and the corresponding overdispersion parameters.

$$T_{i,j,k,c} \perp\!\!\!\perp D_{i,k} \mid \lambda_{i,j,c}, \rho_k$$

4.1.3 Baseline prior specification

As an initial modeling choice, weakly informative priors are assigned to all model parameters.

Random effects:

Random effects are assumed to be independent and identically distributed, sharing a common prior. Each effect is modeled as a zero-centered normal variable with variance σ_b^2 , which is itself assigned an Inverse-Gamma prior (2,1).

$$\begin{aligned} b_{i,j} \mid \sigma_b^2 &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_b^2) \\ \sigma_b^2 &\sim \text{Inverse-Gamma}(2, 1) \end{aligned}$$

Fixed effects:

Fixed effects also share the same prior. They are assumed to be distributed as a normal with zero mean and variance τ^2 , which is specified using a weakly informative prior with Inverse-Gamma distribution (2,1).

$$\begin{aligned} \beta \mid \tau^2 &\sim \mathcal{N}(0, \tau^2 I_p), \\ \tau^2 &\sim \text{Inverse-Gamma}(2, 1) \end{aligned}$$

where p is the dimension of $F(\mathbf{X}_{i,c})$.

Donors:

Donor counts are modeled hierarchically on the log scale. As mentioned above, the parameter α_i represents the true number of donors in experiment i . A normal distribution is assumed for $\log(\alpha_i)$:

$$\begin{aligned} \log(\alpha_i) \mid \mu_\alpha, \sigma_\alpha^2 &\stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_\alpha, \sigma_\alpha^2), \\ \mu_\alpha &\sim \mathcal{N}(\mu_{\alpha_0}, \sigma_{\alpha_0}^2), \\ \sigma_\alpha &\sim \text{half-}\mathcal{N}(0, s_{\alpha_0}) \end{aligned} \tag{8}$$

For the general model the hyperparameters are set to: $\mu_{\alpha_0} = 0$, $\sigma_{\alpha_0}^2 = 10^2$ and $s_{\alpha_0} = 5^2$.

Overdispersion:

$$\begin{aligned} \phi_T &\sim \text{Exp}(\phi_{T_{rate}}) \\ \phi_D &\sim \text{Exp}(\phi_{D_{rate}}) \end{aligned}$$

The exponential prior ensures positivity of the overdispersion parameters while inducing shrinkage toward low dispersion values. Smaller values of ϕ correspond to distributions closer to the Poisson model, whereas larger values allow for increased variability. For the general model the hyperparameters are set to: $\phi_{T_{rate}} = 0.1$ and $\phi_{D_{rate}} = 0.1$

4.2. Extended model specification

To capture potential non-additive effects suggested by the exploratory analysis, the model complexity is increased to test if adding higher terms or interactions between covariates might lead to a better model. The linear predictor using the baseline design matrix $\mathbf{X}$ is

$$b_{i,j} + \beta_0 + \beta_1 [IBU] + \beta_2 [DMSO] + \beta_3 Temp \tag{9}$$

The design matrix X is modified adding the interactions between covariates, $X_{\text{interactions}}$, and the linear predictor becomes

$$b_{i,j} + \beta_0 + \beta_1 [IBU] + \beta_2 [DMSO] + \beta_3 Temp + \beta_4 [IBU] \times [DMSO] + \beta_5 [IBU] \times Temp + \beta_6 [DMSO] \times Temp \quad (10)$$

Lastly, the quadratic terms of the covariates are added to $X_{\text{interactions}}$ and the linear predictor with new design matrix X_{full} becomes

$$b_{i,j} + \beta_0 + \beta_1 [IBU] + \beta_2 [DMSO] + \beta_3 Temp + \beta_4 [IBU] \times [DMSO] + \beta_5 [IBU] \times Temp + \beta_6 [DMSO] \times Temp + \beta_7 [IBU]^2 + \beta_8 [DMSO]^2 + \beta_9 Temp^2 \quad (11)$$

4.3. Shrinkage priors for covariate selection

Increasing the model complexity leads to high variance and a larger number of fixed-effects parameters. Covariate selection is therefore performed to shrink the irrelevant parameters to zero while keeping the important ones. In the Bayesian approach covariate selection is achieved by choosing sparse prior. Different sparse priors are selected and compared:

4.3.1 Lasso

The Bayesian Lasso induces global shrinkage of the regression coefficients, encouraging sparsity by shrinking irrelevant parameters toward zero.

$$\beta \mid \lambda_{\text{Lasso}} \sim \text{Double-Exp}(0, 1/\lambda_{\text{Lasso}}) \\ \lambda_{\text{Lasso}} \sim \text{Half-Cauchy}(0, 1)$$

The parameter λ_{Lasso} controls the overall amount of shrinkage: larger values correspond to stronger penalization and increased sparsity.

4.3.2 Horseshoe

Unlike the Lasso prior, which applies uniform shrinkage, the Horseshoe prior allows coefficient-specific shrinkage through its hierarchical structure.

$$\beta_m \mid \lambda_m, \tau \sim \mathcal{N}(0, \tau^2 \lambda_m^2), \\ \lambda_m \sim \text{Half-Cauchy}(0, 1) \\ \tau \sim \text{Half-Cauchy}(0, 1)$$

The Horseshoe prior combines a global shrinkage parameter τ with local shrinkage parameters λ_m .

4.3.3 Regularized Horseshoe

The regularized Horseshoe introduces the parameter h , which limits the variance of large coefficients.

$$\beta_m \mid \tau, \lambda_m, h^2 \sim \mathcal{N}(0, \tau^2 \tilde{\lambda}_m^2), \\ \tau \sim \text{Half-Cauchy}\left(0, \frac{p_0}{(p - p_0)} \frac{\sigma_\alpha}{\sqrt{N}}\right) \\ \lambda_m \sim \text{Half-Cauchy}(0, 1) \\ h^2 \sim \text{Inverse-Gamma}(0.5, 0.5) \\ \tilde{\lambda}_m = \frac{\sqrt{h^2} \lambda_m}{\sqrt{h^2 + \tau^2 \lambda_m^2}}$$

4.3.4 R2D2

The R2D2 prior controls shrinkage by placing a prior on the model coefficient of determination R^2 , which determines the total explained variance [5].

$$\begin{aligned}\beta_m \mid \sigma^2, R^2, \phi_m &\sim \mathcal{N}\left(0, \sigma^2 \frac{R^2}{1 - R^2} \phi_m\right) \\ \phi &\sim \text{Dirichlet}(a_\pi, \dots, a_\pi) \\ R^2 &\sim \text{Beta}(a, b)\end{aligned}$$

Small values of R^2 imply strong global shrinkage. The Dirichlet prior distributes the total signal across coefficients, allowing sparse solutions when $a_\pi < 1$.

4.3.5 Spike&Slab

The Spike&Slab mixture prior on β_m is approximated by considering a combination of two continuous distributions: a narrow component centered at zero (the spike) and a diffuse component (the slab).

$$\beta_m \mid \tau_{\text{spike}}, \tau_{\text{slab}} \sim \mathcal{N}(0, \tau_{\text{spike}}^2) + \mathcal{N}(0, \tau_{\text{slab}}^2) \quad (12)$$

4.4. Posterior predictive check

Model adequacy is assessed through posterior predictive checks. Specifically transconjugant counts are sampled from the Poisson and Negative Binomial distribution using parameter values drawn from the posterior distributions. These counts are then compared to the observed transconjugant counts in order to evaluate the model's goodness of fit.

4.5. Prior sensitivity analysis

To assess the sensitivity of the model to prior choices, a range of hierarchical prior specifications are tested. For the transconjugant model, the selected priors on the variances of both the random and fixed effects are Inverse-Gamma, Half-Student t and Half-Normal, with different combinations of hyperparameters explored. For the donor model, the hyperparameters μ_{α_0} , $\sigma_{\alpha_0}^2$, and s_{α_0} are varied.

5. Posterior Inference and Model Assessment

In this section all results are illustrated and discussed. STAN is used to define the Bayesian models and perform posterior inference via MCMC sampling.

Table 3: Simulation configurations and computational time, where N denotes the number of configurations and H denotes total hours spent for indicated row. For co-variate selection, there are 2 models (Poisson and Negative Binomial), each with 3 configurations: X , X with interactions, and X with interactions and quadratic terms. However, simulations exceeding 1 hour were dropped, which included all configurations with interactions and quadratic terms. For Sensitivity Analysis, the approach explained in Subsection 4.5 is followed. For NB grid search, we used 6 values between 0.05 and 1.5 for the hyperparameter, resulting in a 6×6 grid search with the NB model and two configurations: X and X with interactions.

Task	Configurations	N.	H.
Co-variate selection (β)	Simple	6	3
	Lasso	6	3
	Horseshoe	4	4
	Regularized Horseshoe ($p_0 = 2$)	4	4
	Regularized Horseshoe ($p_0 = 1$)	4	4
	R2D2	4	4
	Spike & Slab	6	4.5
Sensitivity Analysis	Inverse Gamma	12	6
	Half Normal	6	3
	Half-Student t	4	2
NB grid search	6×6 grid search on $\phi_{D_{rate}}$ and $\phi_{T_{rate}}$	72	40
Total		128	77.5

5.1. Simple Poisson Model:

The general model fixing as likelihood the Poisson distribution is implemented. Convergence is achieved for all parameters and trace plots show good mixing and stable chains, as illustrated in Figure 20a. Posterior credible intervals are computed for the fixed effects parameters β and showed in Figure 7.

The intercept β_0 is centered between -11 and -10 and represents the average log-transconjugation rate per donor when all covariates are set to zero.

The remaining coefficients quantify the effect of the chemical stressors on the transconjugation rate in terms of inactivation. Since the model is specified on the log scale, a positive coefficient corresponds to a reduction in the inactivation effect, i.e. an increase in the transconjugation rate, whereas a negative coefficient indicates stronger inactivation and thus a decrease in the rate relative to the average inactivation. A coefficient whose credible interval includes zero suggests no clear evidence of an inactivation effect.

The coefficient β_1 , associated with IBU, is centered around zero and does not appear to be statistically relevant under this model, as its credible interval includes zero.

In contrast, β_2 , corresponding to DMSO, shows a clear positive effect, since its credible interval does not include zero. This indicates a reduction in the inactivation effect and therefore an increase in the transconjugation rate. The credible interval for β_3 , associated with temperature, is considerably wider and overlaps zero, reflecting greater posterior uncertainty and weaker evidence for a consistent directional effect.

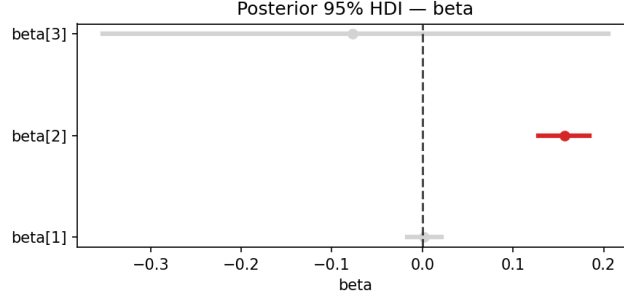


Figure 7: 95% HDI for beta - Simple Poisson Model

In Figure 8 the posterior distribution of the parameters β are plotted.

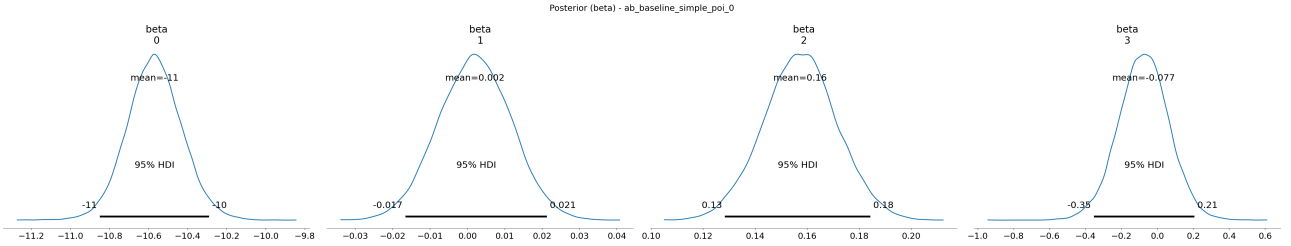


Figure 8: Posterior distribution for beta - Simple Poisson Model

Credible intervals for both α_i and b_{ij} are also computed and showed in Figure 9 and Figure 10, respectively. The posterior credible intervals for the parameters α_i show substantially different means and variances across experiments, justifying the use of a hierarchical Bayesian model for donors, rather than treating donor counts as fixed values.

For most experimental replicates, the intervals for the random effects are centered around zero and largely overlap it. An exception is observed for the replicates of the first experiment and experiment 6, where the posterior intervals are clearly shifted away from zero. This suggests the presence of a specific experimental effect not fully captured by the fixed covariates included in the model.

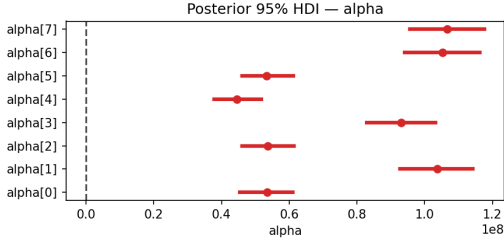


Figure 9: 95% HDI for α - Simple Poisson Model

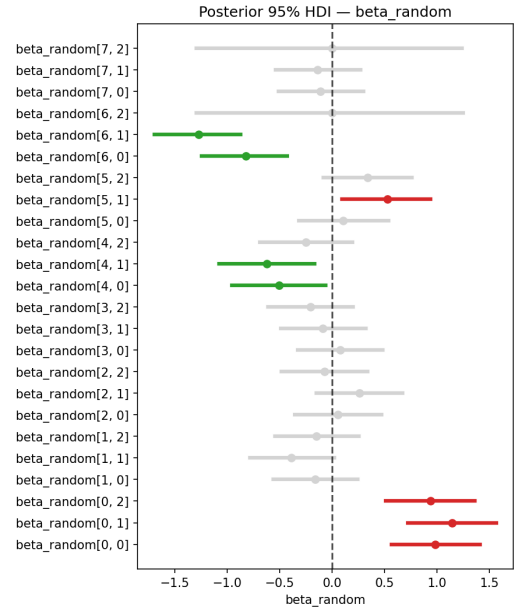


Figure 10: 95% HDI for β_{random} - Simple Poisson Model

Posterior predictive check is performed to evaluate the goodness of the Poisson model. The plot in Figure 11 clearly shows for each trasconjugant count observed, the posterior predictive mean together with the corresponding 95% credible interval obtained from simulated draws of the posterior predictive distribution.

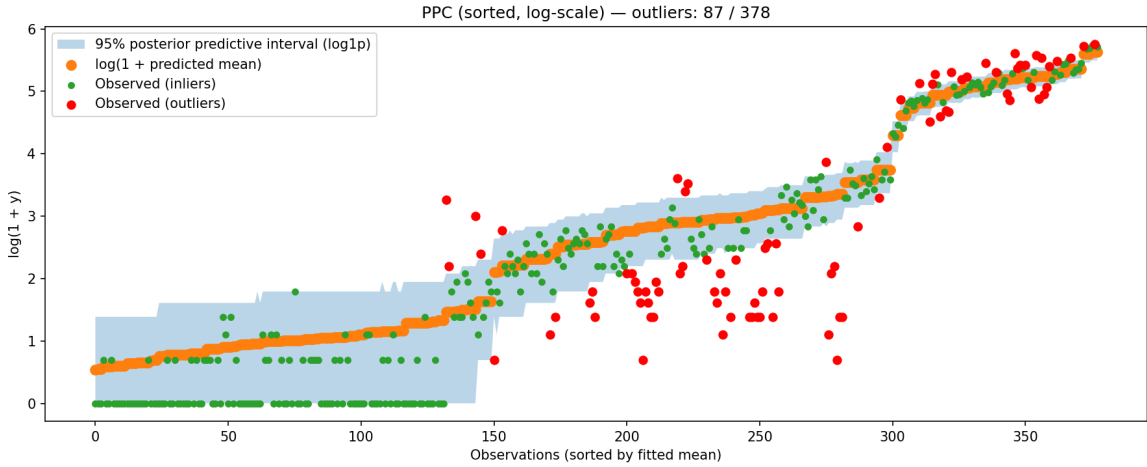


Figure 11: Posterior Predictive Check - Simple Poisson Model

A large number of observations fall outside the 95% credible interval, 87 out 378, which means that they are not explained by the model. This suggests that the Poisson model may be too restrictive and fails to capture the variability of the data. For this reason, the Negative Binomial distribution is introduced.

5.2. Simple Negative Binomial Model:

The same approach as before is followed: the simple model with Negative Binomial distribution is fitted. Convergence is achieved for all parameters and trace plots show stable chains with satisfactory mixing, without evident anomalies, as shown in Figure 26a.

Posterior credible intervals are shown in Figure 12. Differently from the Poisson model, all three intervals overlap

zero. This indicates weaker evidence for a significant effect of the chemical stressors on the transconjugacy rate when overdispersion is taken into account.

The intervals are generally wider than those obtained under the Poisson model, reflecting greater uncertainty in the parameter estimates.

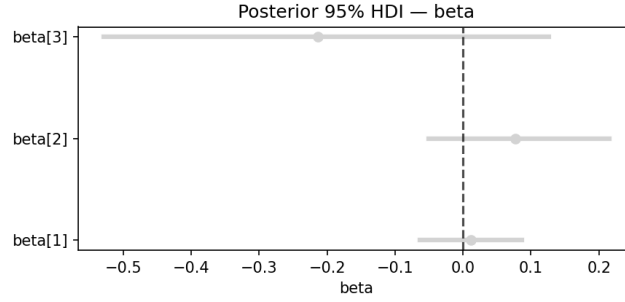


Figure 12: 95% HDI for beta — Simple Negative Binomial Model

In Figure 13 the posterior distribution of β is shown. Compared to the Poisson model, the posterior distributions are more spread out.

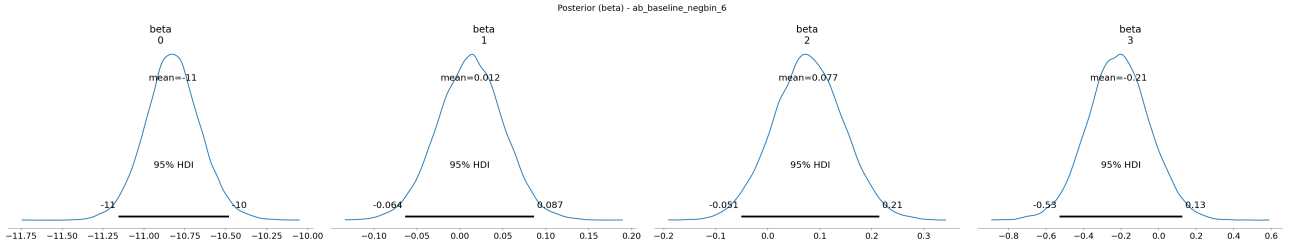


Figure 13: Posterior distribution for beta - Simple Negative Binomial Model

Regarding the estimates of α_i and b_{ij} , a similar behaviour to the previous one is observed, as shown in Figure 14 and Figure 15. Again, a higher uncertainty is observed, as reflected by the wider credible intervals. The introduction of the Negative Binomial distribution does not substantially change the qualitative pattern of parameters α_i and b_{ij} , but it increases the uncertainty associated with their estimates.

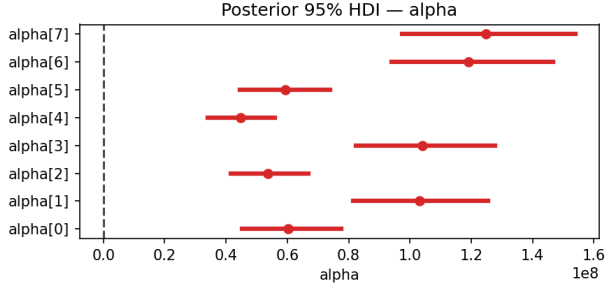


Figure 14: 95% HDI for α — Simple Negative Binomial Model

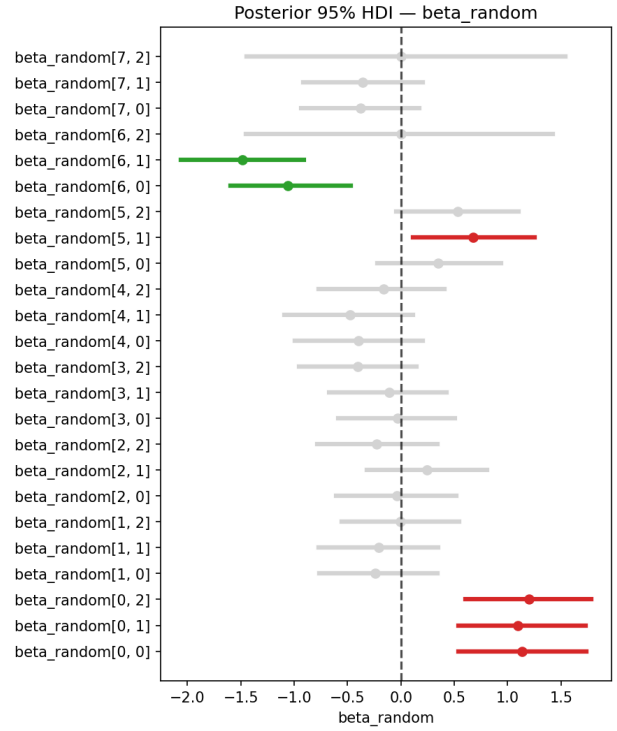


Figure 15: 95% HDI for β_{random} - Simple Negative Binomial Model

The posterior predictive check plot in Figure 16 shows a clear improvement compared to the Poisson model. Only 3 observations fall outside the credible intervals, which are noticeably wider and they are overestimating uncertainty with respect to the Poisson model.

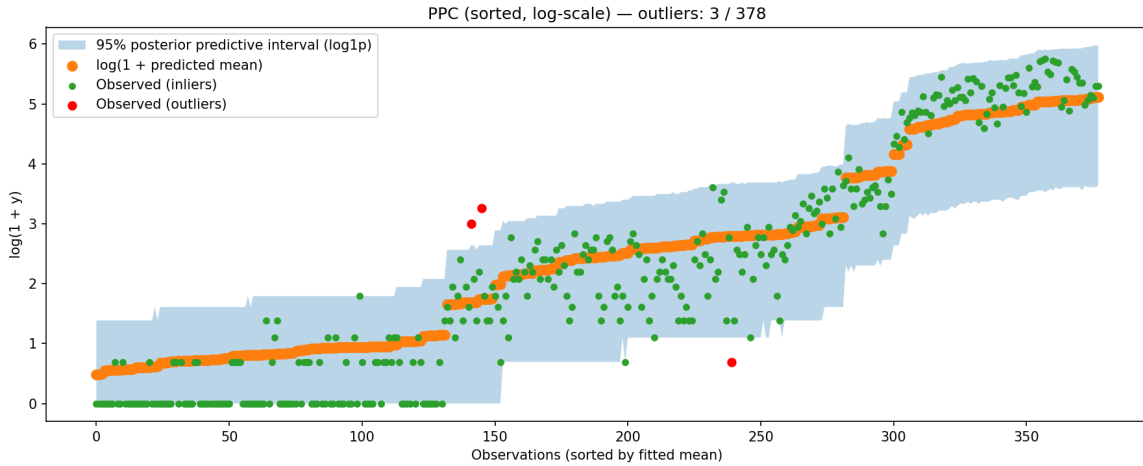


Figure 16: Posterior Predictive Check - Simple Negative Binomial Model

5.3. Shrinkage models comparison

As presented in Subsection 4.2, the model complexity is increased by adding interactions and quadratic terms. Different models are implemented using the sparse priors listed in Subsection 4.3. We select models with divergence smaller than 5% for consistency and correctness of our modeling as presented in Table 4. Models with more than 5% indicate lack of convergence and therefore limited reliability of the corresponding estimates.

Table 4: Divergence rates (%) for Poisson and Negative Binomial regression models across covariate sets (X , X_{full} , $X_{\text{interactions}}$). Only model design configurations with divergence rate $< 5\%$ are reported. The Horseshoe prior assumes $p_0 = 2$ (expected number of non-negligible coefficients).

Base Model	β Prior	X Input	Divergence Rate
Poisson	Simple	X	0.00
Poisson	Simple	$X_{\text{interactions}}$	0.00
Poisson	Lasso	X	0.00
Poisson	Lasso	$X_{\text{interactions}}$	0.00
Poisson	Lasso	X_{full}	0.00
Poisson	Horseshoe	$X_{\text{interactions}}$	1.09
Poisson	Regularized Horseshoe	X	1.05
Poisson	Regularized Horseshoe	$X_{\text{interactions}}$	1.95
Poisson	Regularized Horseshoe	X_{full}	3.43
Poisson	R2D2	X	0.45
Poisson	R2D2	$X_{\text{interactions}}$	0.61
Poisson	Spike&Slab	X	0.00
Poisson	Spike&Slab	$X_{\text{interactions}}$	0.00
Negative Binomial	Simple	X	0.00
Negative Binomial	Simple	$X_{\text{interactions}}$	0.00
Negative Binomial	Lasso	X	0.00
Negative Binomial	Lasso	$X_{\text{interactions}}$	0.00
Negative Binomial	Horseshoe	X	1.50
Negative Binomial	Horseshoe	$X_{\text{interactions}}$	2.60
Negative Binomial	R2D2	X	1.16
Negative Binomial	Spike&Slab	X	0.00
Negative Binomial	Half-Normal	X	0.00

To determine which covariates are selected or, to be more precise, which ones can be regarded as meaningfully impactful, first, we examine the credible intervals (CIs) for each model reported in Table 4. A covariate is treated as *selected* when its posterior CI for the corresponding coefficient does *not* include zero.

As shown in Table 5, the dominant covariate under the design matrix X is DMSO. When we use X augmented with interaction terms ($X_{\text{interactions}}$), the dominant effects are DMSO and the IBU $\times$ Temperature interaction. However, once quadratic terms are included (X_{full}), the conclusions change substantially. In particular, among the (few) quadratic specifications with divergence rate below 5%, the resulting posterior intervals do not support a clear selection of covariates. The added flexibility introduced by quadratic terms increases posterior uncertainty and makes inference less decisive. For this reason and consistently with the diagnostic concerns discussed above we do not emphasize models with quadratic terms in the remainder of the discussion.

For the Negative Binomial models, Table 6 shows that the CI for every covariate includes zero. A reasonable interpretation is that, under the Negative Binomial likelihood, the model can accommodate extra Poisson variability through the dispersion (overdispersion) parameter, thereby reducing the need to attribute variation to covariate effects; consequently, posterior mass for the regression coefficients concentrates more strongly around zero. This pattern may also reflect limited power, collinearity among covariates, or overly strong shrinkage

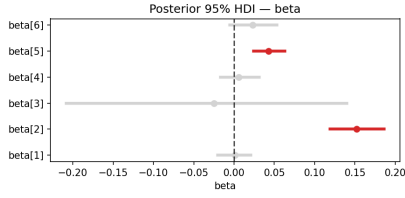
Table 5: Selected Covariates by Model (95% HDI) with Poisson

Model	β Prior	X Input	Intercept	Selected Covariates
Poisson	Simple	X	-10.571 ± 0.280	DMSO (0.157 ± 0.028)
Poisson	Simple	$X_{\text{interactions}}$	-10.568 ± 0.273	DMSO (0.155 ± 0.036), IBU $\times$ Temp (0.044 ± 0.019)
Poisson	Lasso	X	-10.578 ± 0.272	DMSO (0.157 ± 0.028)
Poisson	Lasso	$X_{\text{interactions}}$	-10.577 ± 0.264	DMSO (0.155 ± 0.035), IBU $\times$ Temp (0.044 ± 0.019)
Poisson	Lasso	X_{full}	-10.572 ± 0.273	IBU (-0.203 ± 0.105), IBU $\times$ Temp (0.046 ± 0.019), IBU2 (0.177 ± 0.088)
Poisson	Horseshoe	$X_{\text{interactions}}$	-10.584 ± 0.259	DMSO (0.152 ± 0.034), IBU $\times$ Temp (0.043 ± 0.019)
Poisson	Reg. Horseshoe	X	-10.580 ± 0.278	DMSO (0.155 ± 0.028)
Poisson	Reg. Horseshoe	$X_{\text{interactions}}$	-10.581 ± 0.269	DMSO (0.152 ± 0.033), IBU $\times$ Temp (0.042 ± 0.020)
Poisson	Reg. Horseshoe	X_{full}	-10.580 ± 0.270	IBU $\times$ Temp (0.043 ± 0.020)
Poisson	R2D2	X	-10.577 ± 0.275	DMSO (0.157 ± 0.028)
Poisson	R2D2	$X_{\text{interactions}}$	-10.578 ± 0.272	DMSO (0.154 ± 0.034), IBU $\times$ Temp (0.044 ± 0.019)
Poisson	Spike&Slab	X	-10.537 ± 0.269	DMSO (0.157 ± 0.028)
Poisson	Spike&Slab	$X_{\text{interactions}}$	-10.539 ± 0.265	DMSO (0.157 ± 0.031), IBU $\times$ Temp (0.032 ± 0.024)

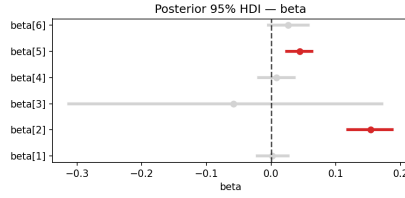
priors. Therefore, we interpret these results as evidence of *weaker support* for covariate effects rather than as definitive evidence of no effect. We ran a grid search on the overdispersion hyperparameters as presented in Table 3, However results still were the same.

Table 6: Selected Covariates by Model (95% HDI) with Negative Binomial

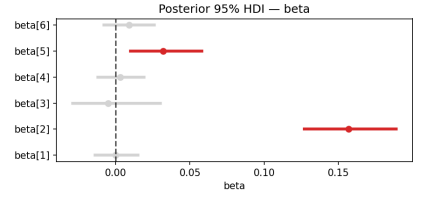
Model	β Prior	X Input	Intercept	Selected Covariates
Negative Binomial	Simple	X	-10.825 ± 0.337	—
Negative Binomial	Simple	$X_{\text{interactions}}$	-10.822 ± 0.306	—
Negative Binomial	Lasso	X	-10.829 ± 0.330	—
Negative Binomial	Lasso	$X_{\text{interactions}}$	-10.829 ± 0.313	—
Negative Binomial	Horseshoe	X	-10.839 ± 0.329	—
Negative Binomial	Horseshoe	$X_{\text{interactions}}$	-10.838 ± 0.314	—
Negative Binomial	R2D2	X	-10.829 ± 0.332	—
Negative Binomial	Spike&Slab	X	-10.764 ± 0.336	—
Negative Binomial	Half-Normal	X	-10.826 ± 0.345	—



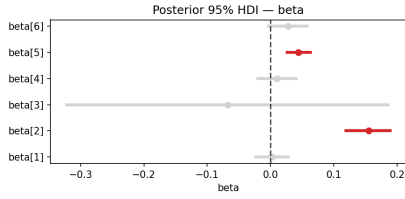
Horseshoe



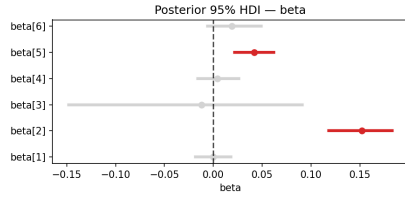
R2D2



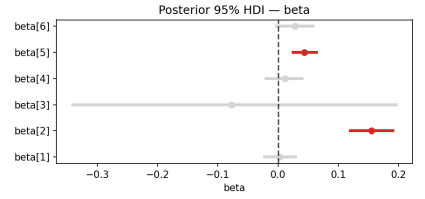
Spike&Slab



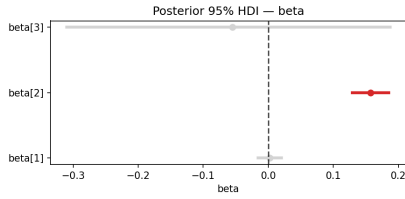
Lasso



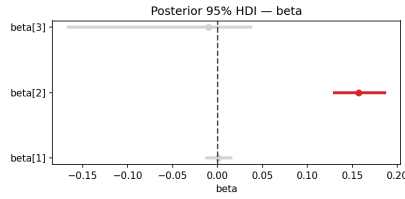
Regularized Horseshoe



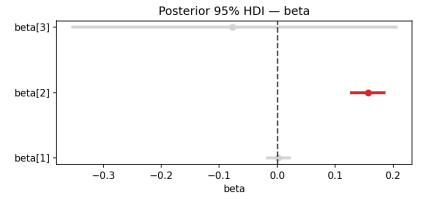
Simple



R2D2



Spike&Slab



Simple

Figure 17: Posterior 95% highest-density intervals (HDIs) for regression coefficients β under the Poisson models considered. Panels compare different β priors (Gaussian, Lasso, Horseshoe, Regularized Horseshoe, R2D2, and Spike&Slab) and two design specifications (X and $X_{\text{interactions}}$). Covariates whose HDIs exclude zero indicate coefficients with stronger posterior support.

5.4. Sensitivity analysis

A total of 20 models are implemented, as described in Subsection 4.5, and all show satisfactory convergence. The results do not highlight any notable differences. The LOO metric, shown in the table below, is stable across models with X and $X_{\text{interactions}}$, as the posterior means of the β parameters in Figure 18. This behavior indicates that the model is robust to the choice of prior specification.

Table 7: LOO comparison across prior specifications

Prior type	Prior params	μ_{α_0}	σ_{α_0}	Covariates	LOO
Inverse-Gamma	shape=3, scale=0.18	0	5	$X_{\text{interactions}}$	-1602.81
Inverse-Gamma	shape=3, scale=0.72	0	5	$X_{\text{interactions}}$	-1602.94
Inverse-Gamma	shape=3, scale=0.18	0	10	$X_{\text{interactions}}$	-1603.13
Half-Student t	df=3, scale=0.5	0	1	$X_{\text{interactions}}$	-1603.23
Half-Normal	$\sigma_b = 0.5$	0	10	$X_{\text{interactions}}$	-1603.32
Inverse-Gamma	shape=3, scale=2.0	0	10	$X_{\text{interactions}}$	-1603.54
Half-Normal	$\sigma_b = 0.5$	0	1	$X_{\text{interactions}}$	-1603.81
Inverse-Gamma	shape=3, scale=0.72	0	10	$X_{\text{interactions}}$	-1603.87
Half-Normal	$\sigma_b = 1$	0	10	$X_{\text{interactions}}$	-1604.00
Half-Student t	df=3, scale=0.5	0	10	$X_{\text{interactions}}$	-1604.24
Inverse-Gamma	shape=3, scale=2.0	0	5	$X_{\text{interactions}}$	-1604.39
Inverse-Gamma	shape=3, scale=0.18	0	5	X	-1608.18
Inverse-Gamma	shape=3, scale=2.0	0	10	X	-1608.52
Half-Normal	$\sigma_b = 1$	0	10	X	-1608.57
Half-Normal	$\sigma_b = 0.5$	0	1	X	-1608.61
Half-Normal	$\sigma_b = 0.5$	0	10	X	-1608.65
Inverse-Gamma	shape=3, scale=0.18	0	10	X	-1608.73
Inverse-Gamma	shape=3, scale=0.72	0	5	X	-1608.80
Inverse-Gamma	shape=3, scale=0.72	0	10	X	-1608.89
Half-Student t	df=3, scale=0.5	0	10	X	-1609.51
Half-Student t	df=3, scale=0.5	0	1	X	-1609.56
Inverse-Gamma	shape=3, scale=2.0	0	5	X	-1610.25

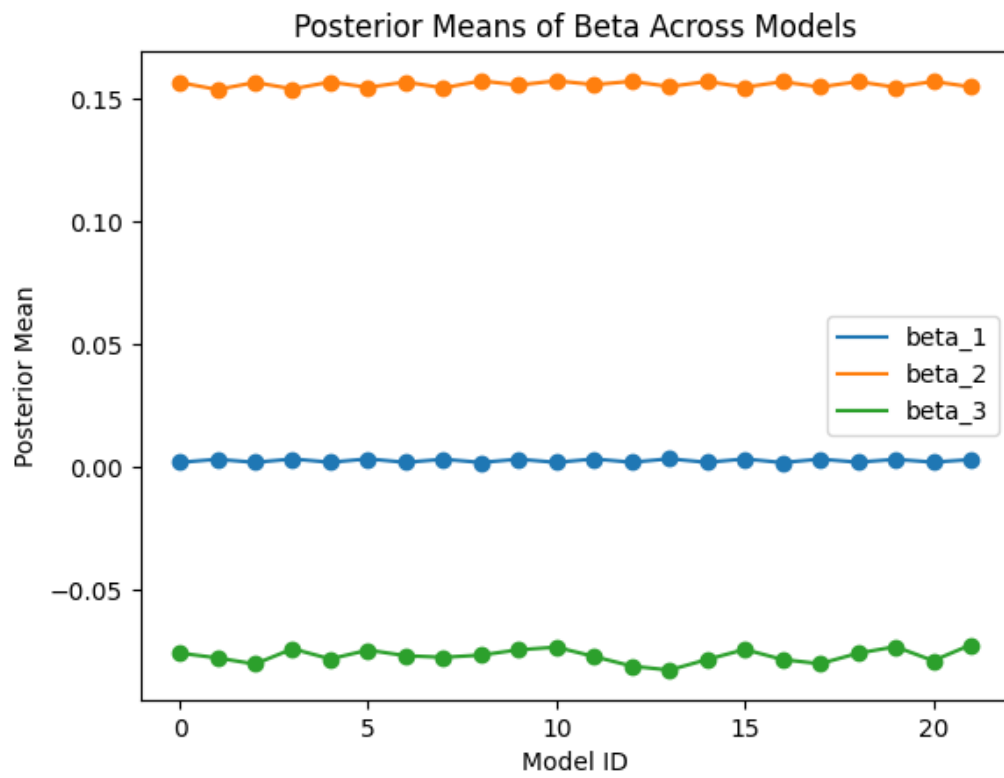


Figure 18: β posterior means across models

6. Conclusions

Among the considered stressors, DMSO shows the most consistent association with the transconjugation rate. In the selected models with interaction terms and shrinkage priors, its coefficient remains positive with credibility intervals that exclude zero. Ibuprofen does not exhibit a clear main effect when considered alone. However, the interaction between ibuprofen and temperature is supported by the models that include interaction terms, suggesting that its effect depends on the underlying value of the temperature. The main effect of temperature alone is characterized by wider credibility intervals, which indicate much greater uncertainty with respect to DMSO.

This suggests that at different values of temperature the effects of the remaining two chemical stressors are different. Indeed, Figure 19 illustrates the impact of the three chemical stressors on the mean log-rate, corresponding to the log-linear predictor excluding the intercept, highlighting differences between low and high temperature conditions. An opposite pattern of the stressors' effects emerges when comparing the two temperature levels. This behaviour is not well captured by the model, and a possible alternative implementation could be to model the response separately for different temperature values.

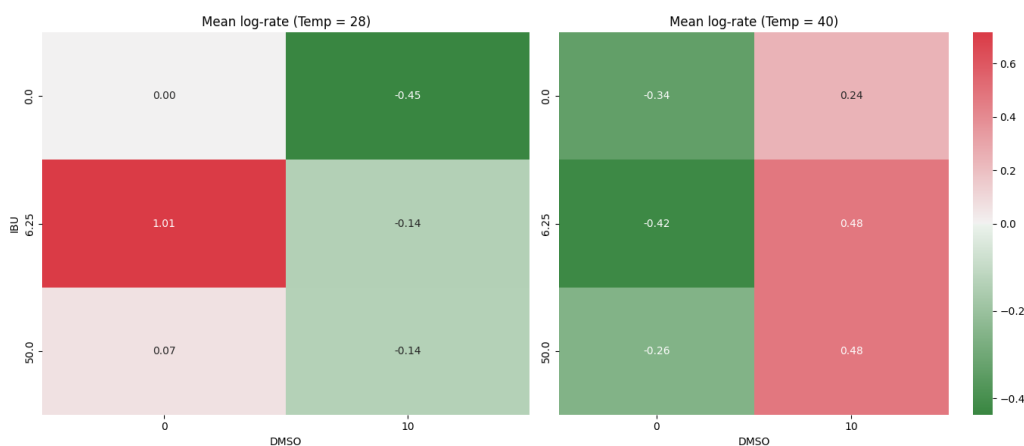


Figure 19: posterior log-mean rate effect DOE

The estimated response surface is not purely additive. Interaction terms improve the description of the experimental structure and capture patterns observed in the exploratory analysis. In particular, the combined presence of DMSO and specific temperature–ibuprofen configurations modifies the log-transconjugation rate beyond the sum of individual effects. Shrinkage priors help identify the components of the response surface that contribute in a significant manner to the fit.

Replicate-level variability is significant. The random effects associated with experiment–replicate combinations account for heterogeneity not explained by the measured covariates.

The joint modeling of donor and transconjugant counts allows uncertainty from dilution factors and counting procedures to propagate into the estimation of the transconjugation rate.

Posterior predictive checks indicate that the Poisson specification underestimates variability, whereas the Negative Binomial likelihood provides a better representation of dispersion in the data.

References

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- [2] Christensen, R., Johnson, W., Branscum, A., Hanson, T. E. (2011). *Bayesian Ideas and Data Analysis*. CRC Press, Boca Raton (USA).
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7. Appendix

7.1. Implementation Details

All models are implemented in Stan [4] via the `CmdStanPy` interface in Python. Computations were performed on a system running Ubuntu 24.04.3, on an Intel Core i7-12650H (8 cores / 16 threads, Alder Lake, 2.69 GHz base), with 8 GB RAM. Implementation code is available https://github.com/ahbagheri01/Progetto_Bayesian_Statistics

7.2. Traces & Autocorrelation Plots

In this section we present some of the models autocorrelation and traces plot.

7.2.1 Simple Poisson

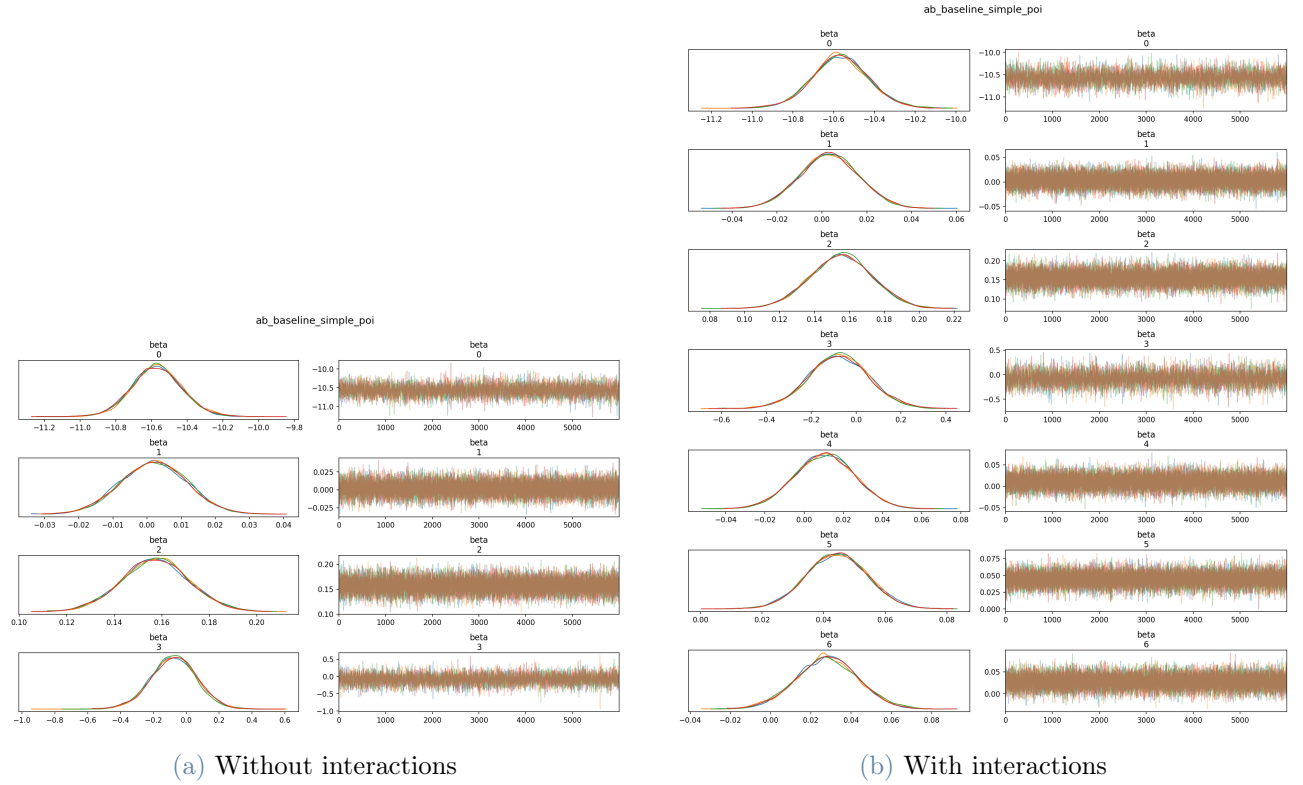
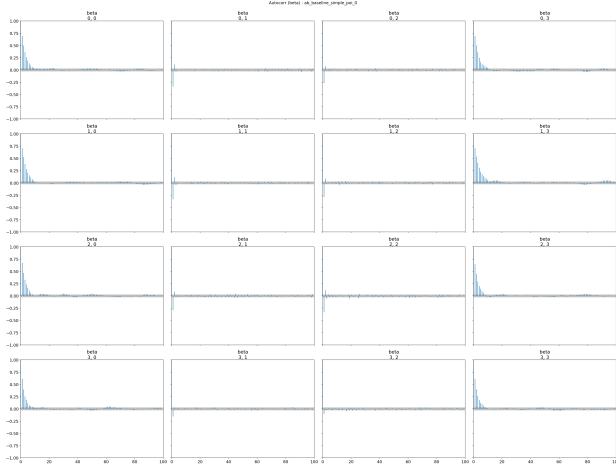
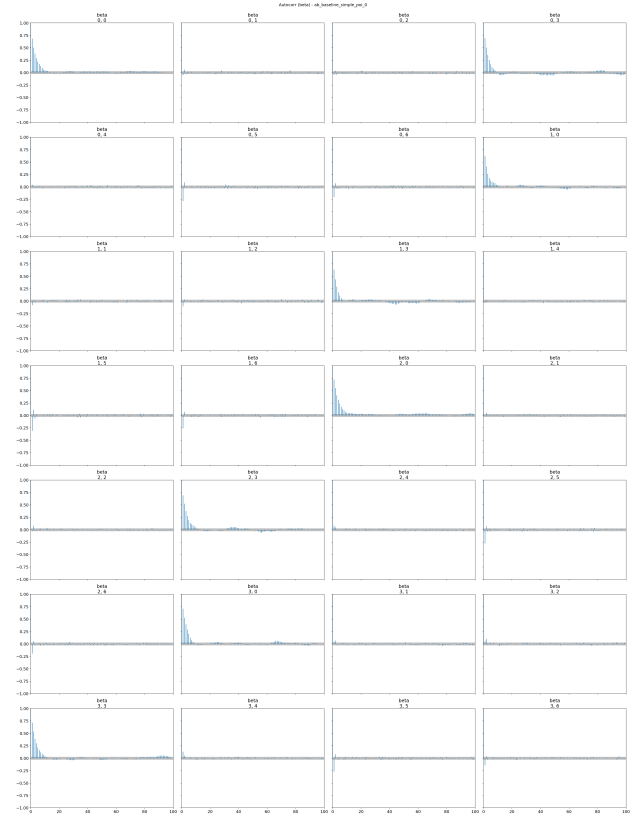


Figure 20: Traces plots for Simple Poisson parameters.



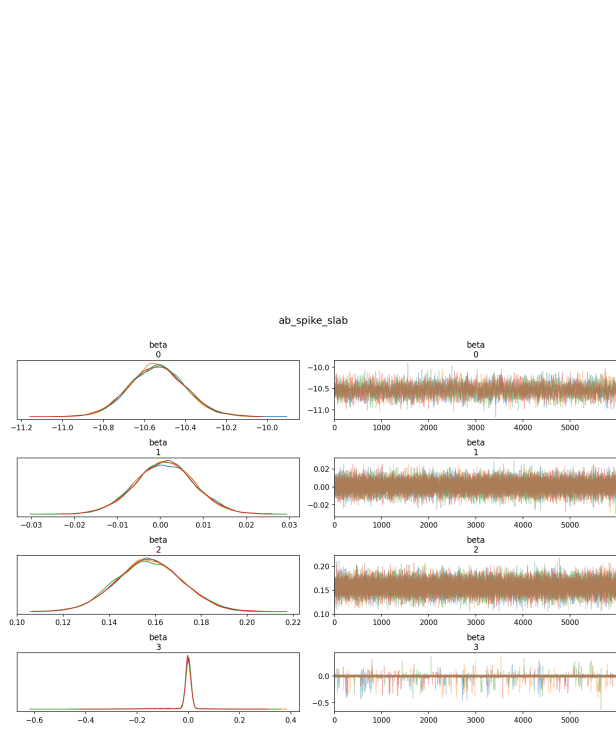
(a) Without interactions



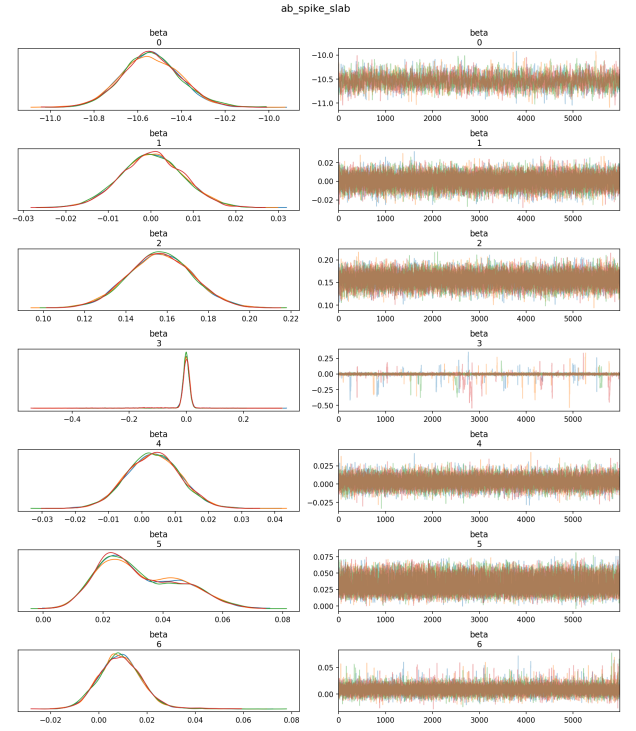
(b) With interactions

Figure 21: Autocorrelation plots for Simple Poisson parameters.

7.2.2 Spike & Slab Poisson

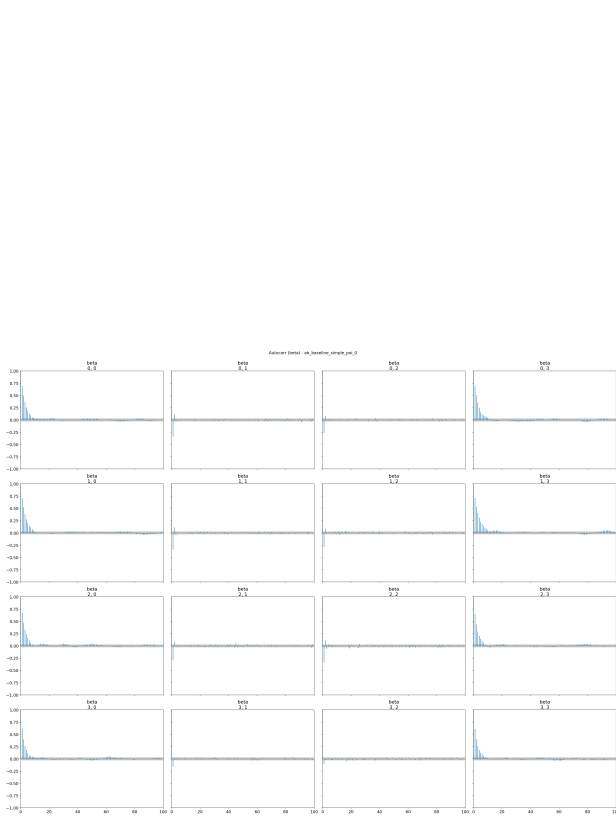


(a) Without interactions

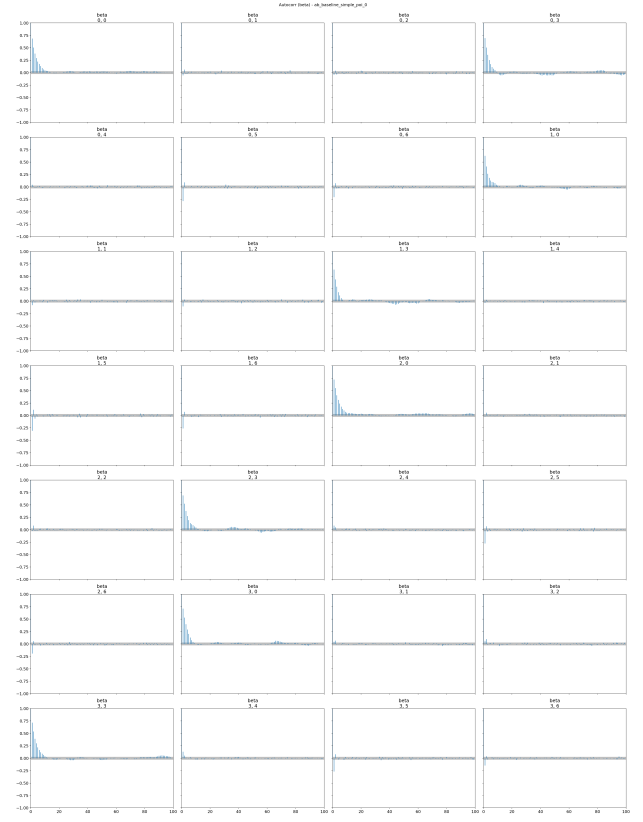


(b) With interactions

Figure 22: Traces plots for Poisson model with sparse prior R2D2 parameters.



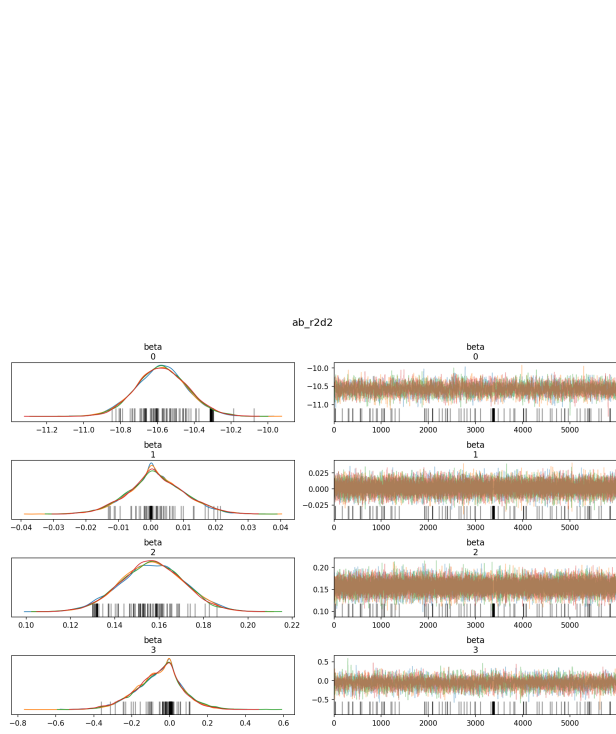
(a) Without interactions



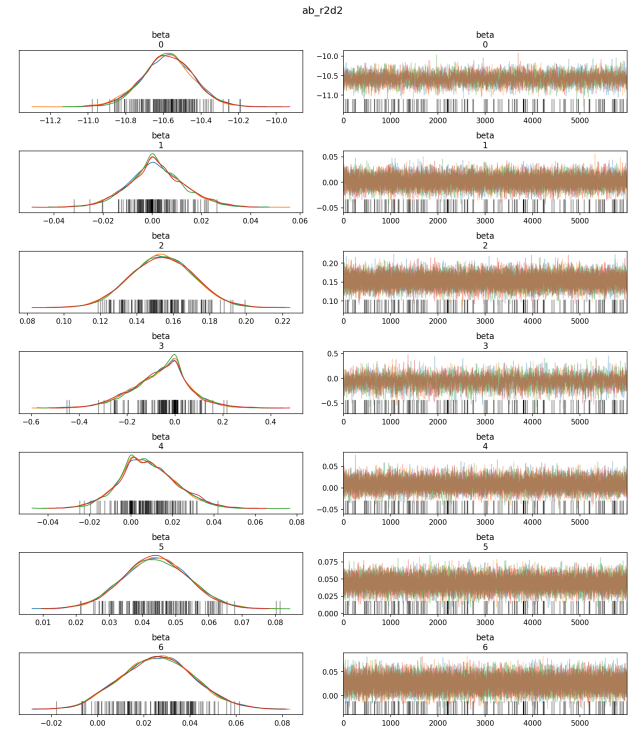
(b) With interactions

Figure 23: Autocorrelation plots for Poisson model with sparse prior Spike & Slap parameters.

7.2.3 R2D2 Poisson

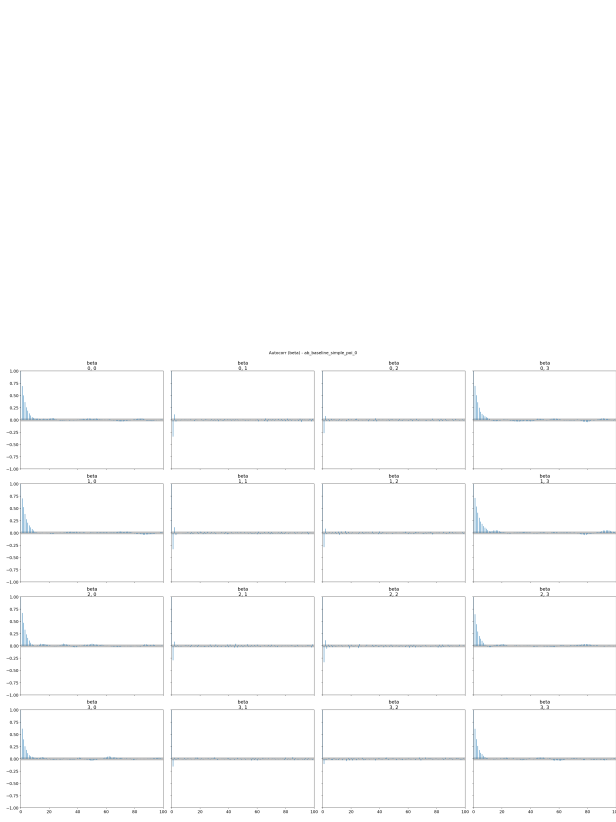


(a) Without interactions

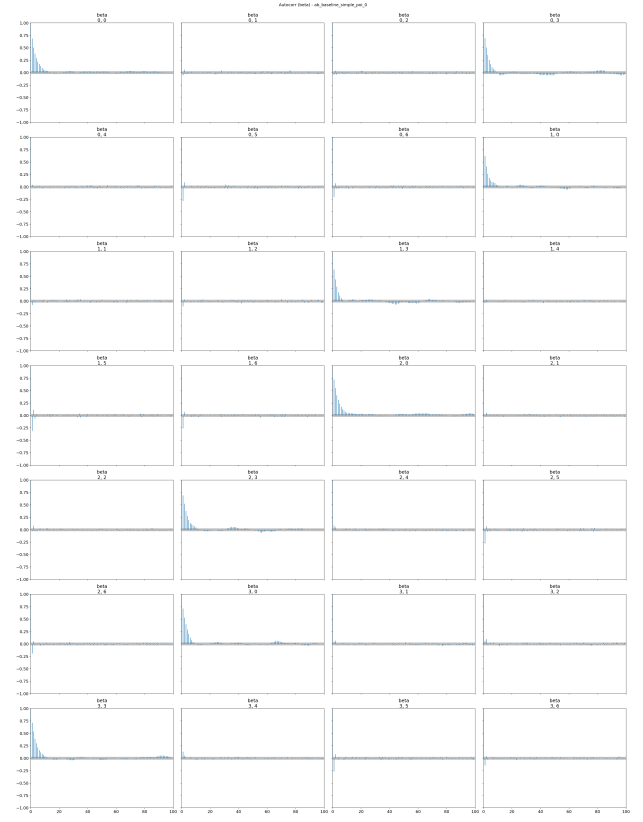


(b) With interactions

Figure 24: Traces plots for Poisson model with sparse prior Spike & Slap parameters.



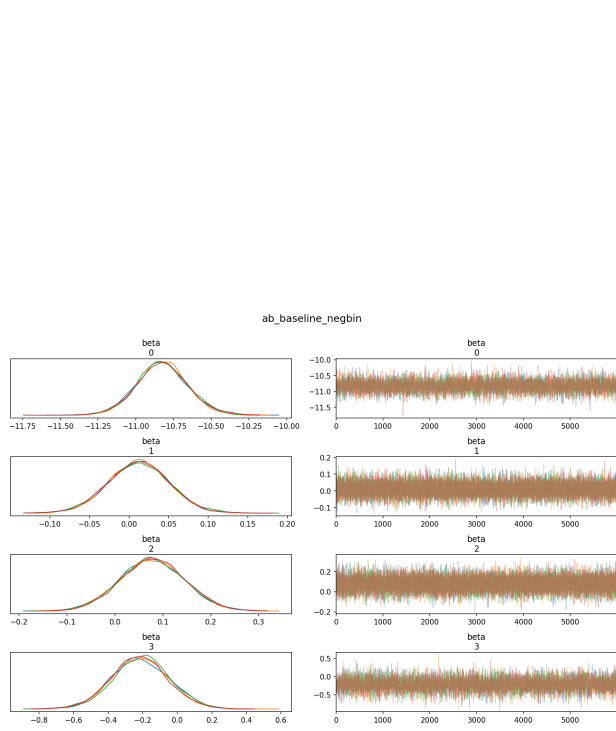
(a) Without interactions



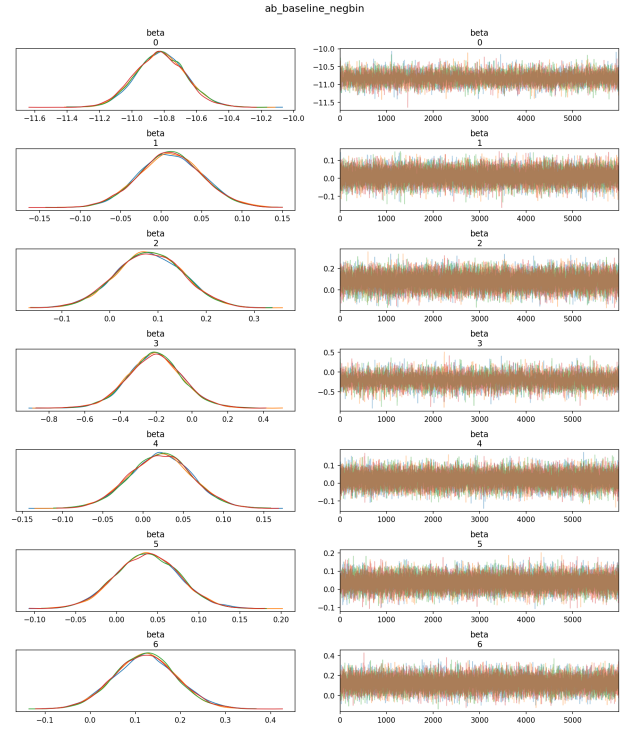
(b) With interactions

Figure 25: Autocorrelation plots for Poisson model with sparse prior Spike & Slap parameters.

7.2.4 Simple Negative Binomial

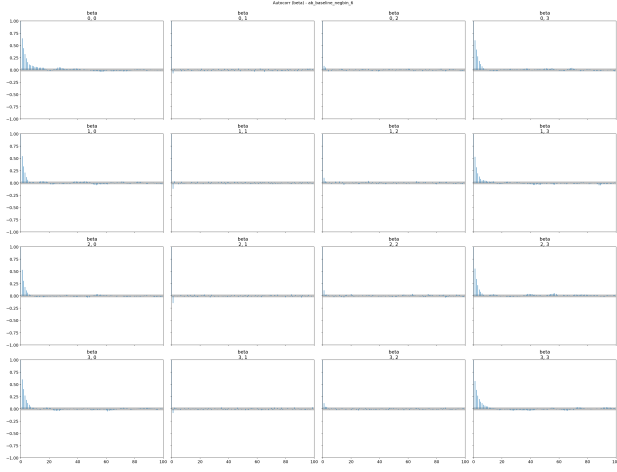


(a) Without interactions

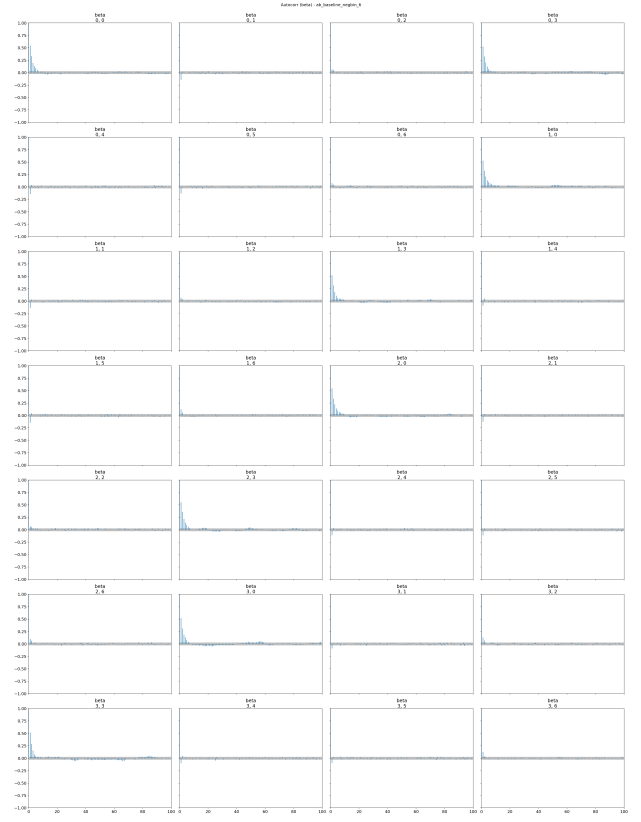


(b) With interactions

Figure 26: Traces plots for Simple Poisson parameters.



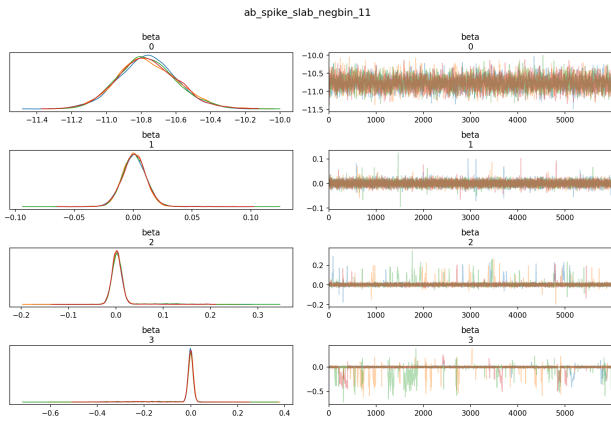
(a) Without interactions



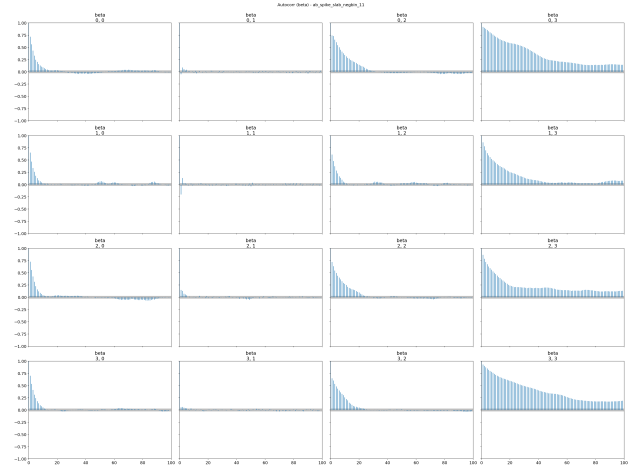
(b) With interactions

Figure 27: Autocorrelation plots for Simple Poisson parameters.

7.2.5 Spike & Slap Negative Binomial



(a) Trace plots



(b) Autocorrelation plots

Figure 28: Autocorrelation and Trace plots for spike & Slap Negative Binomial parameters.

7.2.6 R2D2 Negative Binomial

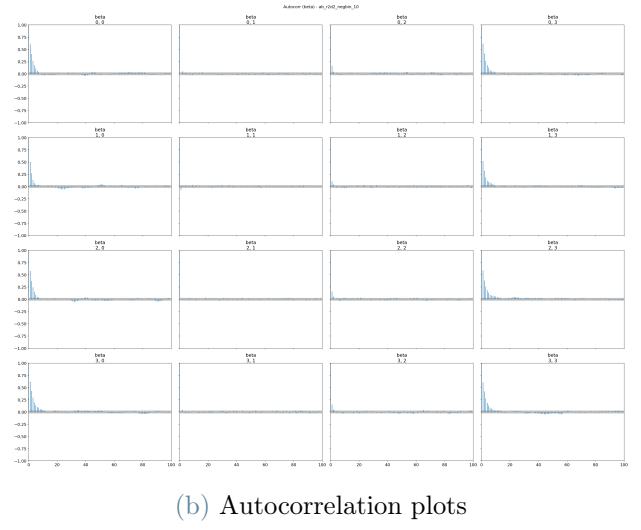
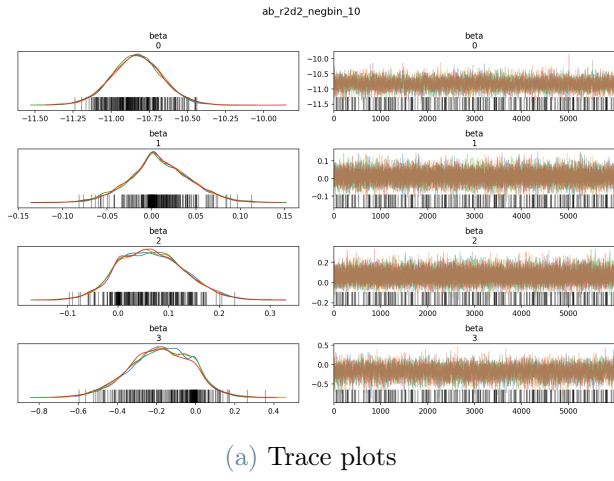


Figure 29: Autocorrelation and Trace plots for R2D2 Negative Binomial parameters.